belonging to seemingly separate species not only having sex but even siring fertile offspring. Most Sapiens living today have about 1–3 per cent Neanderthal DNA,[23] indicating that there once was a child whose father was a Neanderthal and whose mother was a Sapiens (or vice versa). So are Sapiens and Neanderthals the same species or different species? And is 'species' an objective reality that biologists discover, or is it an intersubjective reality that biologists impose?[24]

There are numerous other examples of animals breaking out of their drawers, so the neat bureaucratic division fails to accurately categorise ring species, fusion species and hybrids.[25] Grizzly bears and polar bears sometimes produce pizzly bears and grolar bears.[26] Lions and tigers produce ligers and tigons.[27]

When we shift our attention from mammals and other multicellular organisms to the world of single-cell bacteria and archaea, we discover that anarchy reigns. In a process known as horizontal gene transfer, single-cell organisms routinely exchange genetic material not only with organisms from related species but also with organisms from entirely different genera, kingdoms, orders and even domains. Bacteriologists have a very difficult job keeping tabs on these chimeras.[28]

And when we reach the very edge of life and consider viruses like SARS-CoV-2 (responsible for COVID-19), things become even more complicated. Viruses straddle the supposed rigid boundary between living beings and lifeless matter – between biology and chemistry. Unlike bacteria, viruses aren't single-cell organisms. They aren't cells at all, and don't possess any cellular machinery of their own. Viruses don't eat or metabolise, and cannot reproduce by themselves. They are tiny packets of genetic code, which are able to penetrate cells, hijack their cellular machinery and instruct them to produce more copies of that alien genetic code. The new copies burst out of the cell to infect and hijack more cells, which is how the alien code turns viral. Scientists argue endlessly about whether viruses should count as life-forms or whether they fall outside the boundary of life.[29] But this boundary isn't an objective reality; it is an intersubjective convention. Even if biologists reach a consensus that viruses are life-forms, it wouldn't change anything about how viruses behave; it will only change how humans think about them.